Network Testbed Sprint 2

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Agenda

01 Project Overview

02 Sprint 2 Goals

03 Sprint 2

04 Other Contributions

05 Next Steps

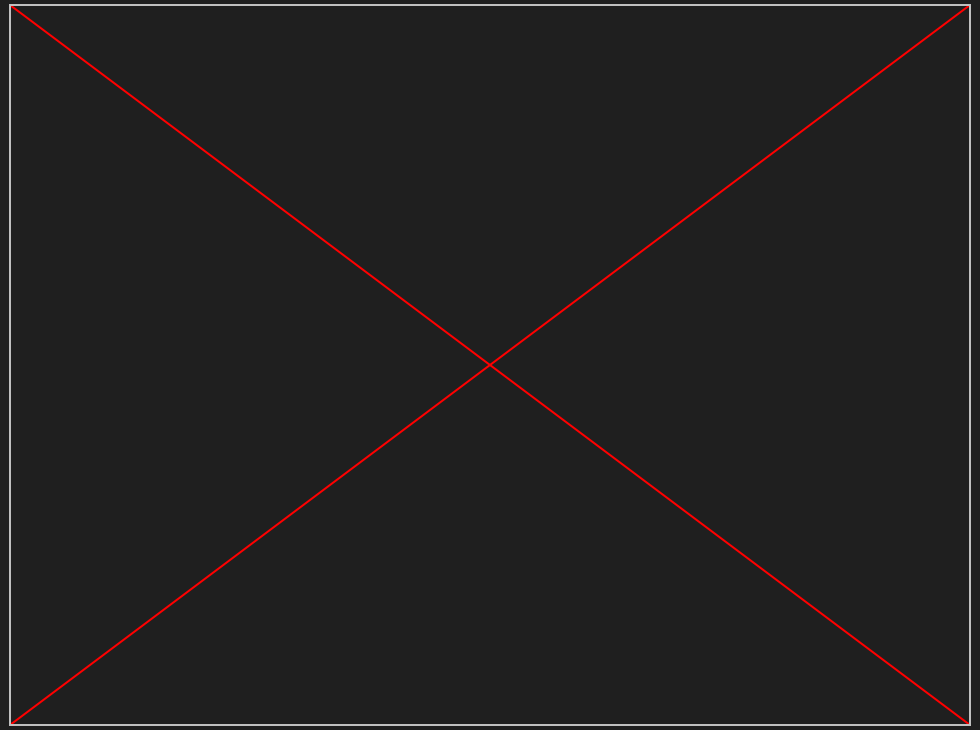
Project Overview

A network simulation platform that represents network traffic during cyberattack scenarios. Designed for open use, we include tested, representative topologies that allow users to test custom network metrics and measure performance under simulated conditions and attacks

Sprint 2 Goals

- Create a basic ui that imports and shows topologies - Completed
- Reflect modifications from ui to Mininet—Completed

GUI Progress - Fernando



Packet Traffic Generator - Dylan

- D-ITG, Distributed Internet Traffic Generator
  - Produce traffic at packet level accurately replicating appropriate stochastic processes.
  - Capable to generate traffic at network, transport, and application layer.
  - Records metrics like throughput, delay, jitter, and packet loss at the receiving end.

Example

clinic1 ITGRecv &
clinic2 ITGRecv &

voice1 ITGSend -a 10.20.10.11 -T UDP -C 50 -c 100 -t 5000 -l voice_to_clinic.log

ITGDec voice_to_clinic.log

```
dylan@dylan-VirtualBox: ~/capstone/tests/urbhos
dylan@dylan-VirtualBox:~/capstone/tests/urbhos$ ITGDec voice_to_clinic.log
ITGDec version 2.8.1 (r1023)
Compile-time options: sctp dccp bursty multiport
---------------------------------------------------------
Flow number: 1
From 10.10.40.41:43885
To    10.20.10.11:8999
---------------------------------------------------------
Total time                =     4.986492 s
Total packets             =          231
Minimum delay             =     0.000000 s
Maximum delay             =     0.000000 s
Average delay             =     0.000000 s
Average jitter            =     0.000000 s
Delay standard deviation =     0.000000 s
Bytes received            =        23100
Average bitrate           =    37.060122 Kbit/s
Average packet rate       =    46.325152 pkt/s
Packets dropped           =            0 (0.00 %)
Average loss-burst size   =     0.000000 pkt
---------------------------------------------------------


*****************  TOTAL RESULTS   ******************

Number of flows           =            1
Total time                =     4.986492 s
Total packets             =          231
Minimum delay             =     0.000000 s
Maximum delay             =     0.000000 s
Average delay             =     0.000000 s
Average jitter            =     0.000000 s
Delay standard deviation =     0.000000 s
Bytes received            =        23100
Average bitrate           =    37.060122 Kbit/s
Average packet rate       =    46.325152 pkt/s
Packets dropped           =            0 (0.00 %)
Average loss-burst size   =            0 pkt
Error lines               =            0
---------------------------------------------------------
dylan@dylan-VirtualBox:~/capstone/tests/urbhos$
```

How this helps us

- Lets us see whether the topologies are actually behaving the way they are configured.
- Lets us test the delays and jitter of link parameters set up. Example (clients, admins, servers)
- Simulating realistic traffic types.
- Testing attacks types such as DoS.

Other Contributions – Selim

- Researched and identified relevant attack scenarios for scripting
- Collaborated with Dylan to evaluate and ensure script compatibility with D-ITG

Other Contributions - Logan

- (ADDITION) Created a script to run the urban topology
  - Assign interfaces and pathways automatically
- Create documentation for urban topology script
- Help with attack scripts

Other Contributions - Paige

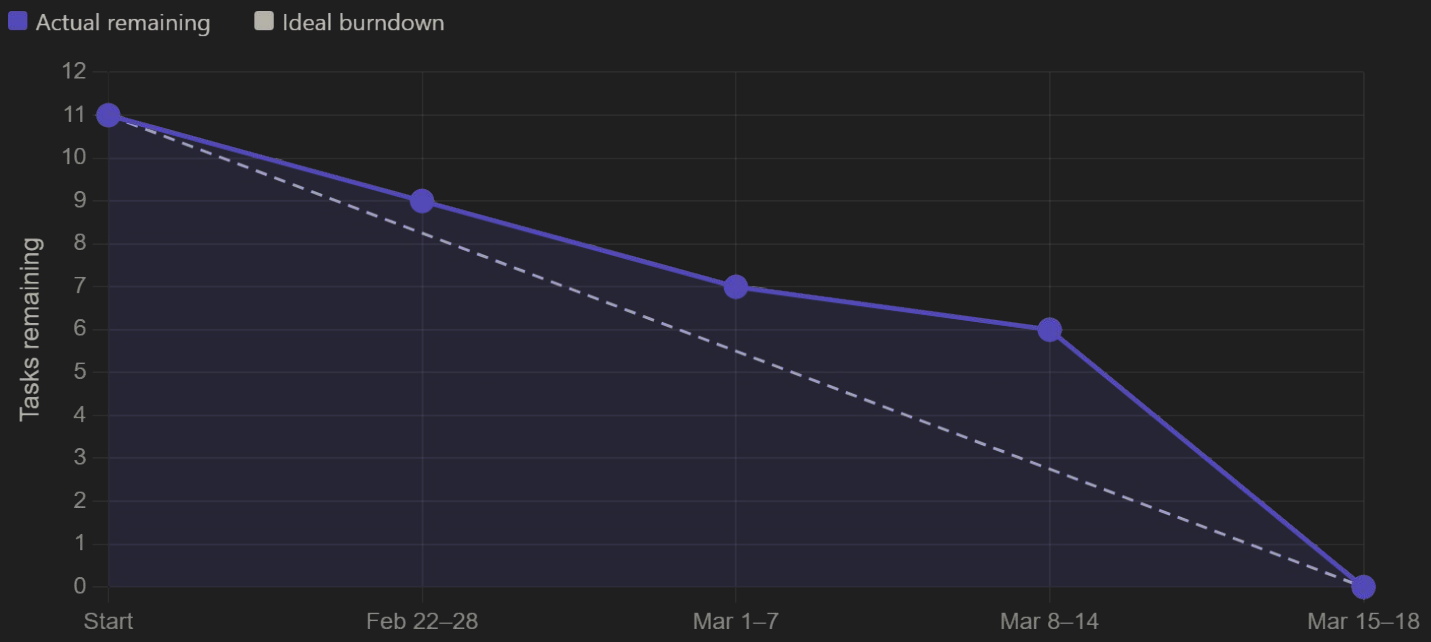
- Helped convert scripts to json
- Finished documentation for the rural-to-urban network

Average Baseline Hours Logged on Clockify

Week	Total hours
Feb 22 - Feb 28	7.1
Mar 1 - Mar 7	8.3
Mar 8 - Mar 14	3.6
Mar 15 - Mar 19	6.3

Burndown chart

Total tasks	Completed	Remaining	Duration
11	11	0	4 wks



Next Steps

Sprint 3 - 04/02

- Incorporate a traffic generator into GUI
- Create network attack scripts and incorporate into UI
  - DDoS, arp spoofing, eavesdropping, etc.

Sprint 4 - 04/16

- Test all attacks on created networks
- Incorporate attack scripts to UI
- Track findings

Thank you!